

ANOVA Assumptions Assessment for Box-Cox Transformed Habitat Features

Taxa Assemblage Groups Analysis

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1 Main Results Table

Table 1: Assessment of ANOVA Assumptions for Box-Cox Transformed Environmental Features

Feature	λ	Shapiro-W	p-value	Normality	Levene's	p-value	Homog.
Temperature (°C)	1.902	0.812	0.000	FAIL	7.978	0.000	FAIL
Depth (m)	0.239	0.986	0.642	PASS	0.970	0.412	PASS
DO Bottom (mg/L)	2.611	0.907	0.000	FAIL	0.656	0.582	PASS
LOI (%)	0.060	0.993	0.958	PASS	2.657	0.056	PASS
MPS (Phi)	3.304	0.920	0.000	FAIL	0.611	0.610	PASS

2 Summary Statistics

Table 2: Summary of Assumption Test Results: Box-Cox vs Original

Metric	Original	Box-Cox
Total features tested	5	5
Normality assumption passed	0	2
Homogeneity assumption passed	3	4
All assumptions passed	0	2

3 Detailed Comparison by Feature

Table 3: Feature-by-Feature Comparison: Original vs Box-Cox Transformed

Feature	Original	Box-Cox	Result
Temperature (°C)	FAIL	FAIL	Same
Depth (m)	FAIL	PASS	Improved
DO Bottom (mg/L)	FAIL	FAIL	Same
LOI (%)	FAIL	PASS	Improved
MPS (Phi)	FAIL	FAIL	Same

4 Box-Cox Transformation Details

The Box-Cox transformation is defined as:

$$y(\lambda) = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(x) & \text{if } \lambda = 0 \end{cases}$$

4.1 Optimal Lambda Values and Interpretations

- **Temperature (°C):** $\lambda = 1.9019$ — Custom power transformation
- **Measured Depth (m):** $\lambda = 0.2392$ — Custom power transformation
- **Water DO Bottom (mg/L):** $\lambda = 2.6106$ — Custom power transformation
- **LOI (%):** $\lambda = 0.0596$ — $\approx$ Log transformation
- **MPS (Phi):** $\lambda = 3.3036$ (shifted by 2.44) — Custom power transformation

5 Methodology Notes

- **Normality:** Assessed using Shapiro-Wilk test on ANOVA residuals ($p > 0.05$ indicates normality)
- **Homogeneity of Variance:** Assessed using Levene's test ($p > 0.05$ indicates homogeneous variances)
- **Independence:** Assumed based on spatial separation of sampling sites
- **Box-Cox Method:** Automatically finds optimal λ to maximize normality of the transformed data
- **Shifting:** Features with non-positive values were shifted by $|\min| + 1$ before transformation
- **Groups:** 4 taxa assemblage groups identified by Ward's hierarchical clustering